



Embodied AI researcher & simulation engineer specialising in motor control and robotics using RL policies grounded in neuromechanics.

Work Experience

2025 - **Postdoctoral Researcher, Imperial College/Meta Wearable Neural Interfaces Research Centre:** Developed neuromechanical models of the hand, and its control policies using RL. Built experimentally validated, closed-loop neural signal synthesis pipeline in massively parallelized simulated environments. Achieved 100x speedup (100k+ sps) by parallelizing via **JAX** and **Warp** on **HPC** clusters.

Led Imperial's contributions on multiple >€500k ERC **grant applications** in robotics and sensor manufacturing projects that met excellence requirements. Developed and iterated on a robotic hand's digital twin from **Fusion360**, **Blender** and **Python** to generate **MJCFs** and extended the simulation with a prototype arm's model. Developed **robotic hand simulation environment** controlled by users in real-time VR to test human-machine interfacing devices. Co-designed **Unity-based image data synthesis** pipeline for object detection and **hand preshaping** system for robotic grasping.

2023-25 **Contractor, Google DeepMind:** Contributed **bug-fixes and PRs** to MuJoCo and related open-source projects and provided community moderation. **Developed features** for the **Unity** plugin to the **MuJoCo** physics engine. Contributed a **system identified compliant robotic end-effector** model to MuJoCo Menagerie, creating multiple levels of detail variants.

2023 **Research Intern, Artanim Foundation:** Investigated social spacing in virtual environments. Developed tools and processes for physics-based animation and locomotion simulation. Implemented interactive environments in VR using Unity. Planning and performing full-body, **multi-actor motion capture sessions** for motion-tracking datasets.

2017-24 **Teaching assistant:** Leading seminars and tutorials, creating and presenting lecture content, planning and grading coursework. Lead TA for Human Neuromechanical Control and Learning at ICL.

Education & Research Experience

2020-24 **PhD project, Imperial College London (Bioengineering Research):** Learnt **locomotion** using intent-driven, **compliant robotic** legs in simulated **RL** environments. Agents trained to maintain dynamic balance on **challenging terrain**, with realistic torque limits. Collected experimental kinematic and kinetics datasets and processed them with TCNs to extract high-level user intent and dynamics. **5 first-author articles published. 8 second-author articles, including in Nature Communications. 1st place presentation in category at iCBEI22.**

Co-first author of two **NeurIPS** competitions (MyoChallenge24/25), designed RL environments (including **bimanual manipulation**) and reward-hacking resistant evaluation metrics (partially through **domain randomisation**) for 100+ competing teams. **Published in NeurIPS 2025, 2026, and ICORR 2025.** Developed and implemented features to utilise hardware accelerated parallelised simulation of these open-sourced



environments. Contributed to procedural programmatic model composition pipeline using the Python **MjSpec** API of MuJoCo.

2019-20 **MEng Biomedical Engineering, Imperial College London:** Projects included building accessible UI for controlling a robotic arm, and CNN-based image processing. Built a data loading, systematic training and documentation systems for my thesis project; EMG based **kinematics prediction** and **activity classifier**. Projects presented at AAATE18, BioMedEng19/22 conferences. **First class honours**, Dean's list.

Skills

- **Simulation & RL:** **MuJoCo, JAX, Warp**, Brax, IsaacLab, Unity (C#), GPU-parallelisation, reward shaping, domain randomization, scene creation via Unity, Blender, converting between URDF and MJCF. Familiarity with physics engines e.g. Genesis, Bullet, SimBody and PhysX.
- **Technical skills:** Signal processing, human-machine interfacing, supervised and reinforcement learning, information theory, gait analysis, motion capture, optimal control, system-identification
- **IT:** Python, C#, Git, C++, ROS, CI/CD pipelines, unit testing
- **Entrepreneurship:** Trained on managing and protecting IP, regulations and tech transfer. Worked with **startups/companies** at different stages (e.g. Neubond, Artanim), making simulations and visualisations they used for fundraising and outreach.
- **Communication:** Experience as teaching assistant, PhD student society chair. **Designed and led three-day workshops on simulating robotics at Summer School of Neurorehabilitation 2023-26.**
- **Supervision:** Contributed to and advised student projects involving **exoskeleton control, teleoperation, tactile sensing**, IMU based motion capture, uncertainty modelling. I supervised a student adapting my framework for **vision-based control simulation of a Unitree H1 over varying terrain**.

Other Interests and Successes

- **Released an open-source [RL-based locomotion learning toolkit](#), [written interactive tutorials](#) and a [robotics modelling utility](#).**
- Contributed peer reviews to multiple journals including Science Robotics, multiple IEEE journals and Wearable Technologies.
- Personal Projects: Arduino and Raspberry Pi electronics projects, programming video games, image processing apps. Organising chair of the CDT Prosthetics & Orthotics Conference 2021.

Key publications

Hodossy, B.K. and Farina, D., 2023. Shared Autonomy Locomotion Synthesis With a Virtual Powered Prosthetic Ankle. IEEE Transactions on Neural Systems and Rehabilitation Engineering

Wang, H., Tan, C.K., Hodossy, B., Lyu, S., Schumacher, P., et al., 2026. MyoChallenge 2024: A New Benchmark for Physiological Dexterity and Agility in Bionic Humans. Advances in Neural Information Processing Systems, 38.

Hodossy, B.K., Crotti, M., Pace, A., Catalano, M.G., Aszmann, O.C., Bicchi, A. and Farina, D., 2026. Towards a Virtual Gait Lab: Testing Prosthetics with Dynamically Simulated Users. IEEE Transactions on Medical Robotics and Bionics.